Study: Numerical modelling of Wave-vegetation interactions

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Abstract

This preliminary study investigates the influence of submerged coastal vegetation on wave propagation and attenuation. Vegetation is represented as an array of rigid vertical cylinders, while the associated energy dissipation is evaluated from the hydrodynamic drag predicted by Morison's equation.

Two numerical models are developed. The first considers a regular wave and combines linear wave theory with an energy-flux balance to predict the evolution of wave height through the vegetated region. The second extends the methodology to an irregular sea state represented by a discretised JONSWAP spectrum. The total nonlinear drag is calculated from the reconstructed velocity field, after which the energy and amplitude of the spectral components are updated progressively along the propagation direction.

The study establishes a preliminary numerical framework for investigating wave–vegetation interactions. The model is not yet fully validated, and further work is required to assess its assumptions, numerical convergence and predictive accuracy.

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1 Introduction

1.1 Objectives

The objective of this project is to investigate the influence of submerged coastal vegetation on wave propagation and attenuation through numerical modelling.

The study begins with a simplified wave-energy model based on linear wave theory and Morison's equation, representing vegetation as an array of rigid vertical cylinders. The aim is to predict wave-height attenuation and investigate the influence of vegetation properties and wave conditions.

2 First Model: Wave Attenuation through Rigid Cylinders under Regular Waves

A simplified model was first developed to represent wave-vegetation interactions.

The vegetation is represented as an array of rigid vertical cylinders. Wave attenuation is calculated from the energy dissipated by hydrodynamic drag, using Morison's equation and the wave energy balance.

The model predicts the evolution of wave height $H(x)$ along the vegetated area.

2.1 Main Assumptions

Wave assumptions:

- Regular, progressive waves propagating along the x-axis.
- Linear wave theory (Airy).
- Constant water depth $h = 5\text{m}$.
- Constant wave period $T = 4.0\text{s}$.
- Wave breaking and reflection are neglected.

Vegetation assumptions:

- Vegetation is represented as rigid, fixed, vertical cylinders.
- Vegetation is present between $x = 0\text{m}$ and $x = 50\text{m}$.
- All cylinders have the same diameter $d = 0.1\text{m}$ and height $h_v = 1\text{m}$.
- Constant drag coefficient $C_D = 1.2$ and inertia coefficient $C_M = 1.0$.
- Cylinders are uniformly distributed, with a density $n = 100$ $(1/\text{m}^2)$.
- Interactions between neighbouring cylinders are not explicitly resolved.

2.2 Linear Wave Theory

The wave number is obtained from the linear dispersion relation:

$$\omega^2 = gk \tanh(kh) \tag{2.1}$$

where:

- $\omega = 2\pi/T$ is the angular wave frequency (rad/s).
- g is the gravitational acceleration (m/s^2).

- $k = 2\pi/L$ is the wave number (rad/m).
- h is the water depth (m).
- L is the wavelength (m).

The free-surface elevation of the regular wave is written as:

$$\eta(x,t) = \frac{H(x)}{2}\cos(kx - \omega t) \tag{2.2}$$

The horizontal orbital velocity is:

$$u(x,z,t) = \frac{H(x)}{2}\omega\frac{\cosh(k(z+h))}{\sinh(kh)}\cos(kx - \omega t) \tag{2.3}$$

The horizontal orbital acceleration is:

$$\frac{\partial u}{\partial t}(x,z,t) = \frac{H(x)}{2}\omega^2\frac{\cosh(k(z+h))}{\sinh(kh)}\sin(kx - \omega t). \tag{2.4}$$

where:

- $H(x)$ is the local wave height (m).
- x is the horizontal position along the wave propagation direction (m).
- z is the vertical coordinate, with $z = 0$ at the still water level (m).
- t is time (s).
- u is the horizontal orbital velocity (m/s).
- $\partial u/\partial t$ is the local horizontal acceleration (m/s^2).

2.3 Morison's Equation and Fluid Force on the Vegetation

Morison's equation is used to calculate the horizontal hydrodynamic force exerted by the fluid on a rigid vertical cylinder.

The force per unit length of cylinder is:

$$f_M = \underbrace{\frac{1}{2}\rho_w C_D d|u|u}_{\text{Drag force}} + \underbrace{\rho_w C_M \frac{\pi d^2}{4}\frac{\partial u}{\partial t}}_{\text{Inertia force}} \tag{2.5}$$

where:

- ρ_w is the water density (kg/m^3).
- C_D is the drag coefficient (-).
- C_M is the inertia coefficient (-).

- d is the cylinder diameter (m).
- u is the horizontal fluid velocity (m/s).
- $\partial u/\partial t$ is the horizontal fluid acceleration (m/s^2).
- f_M is the hydrodynamic force per unit cylinder length (N/m).

Morison's equation gives the force applied by the fluid on one cylinder per unit length. We can then obtain the force applied to the vegetated region per unit volume by multiplying by the cylinder density n:

$$f_M^{vol} = \frac{1}{2}\rho_w C_D n d|u|u + \rho_w C_M n \frac{\pi d^2}{4}\frac{\partial u}{\partial t} \tag{2.6}$$

where n is the number of cylinders per unit horizontal bed area (stems/m^2), and f_M^{vol} is expressed in N/m^3.

Using Newton's third law, the force applied by the vegetation on the fluid is equal and opposite to the force applied by the fluid on the vegetation:

$$f_{\text{veg}\to\text{fluid}} = -f_M^{vol} \tag{2.7}$$

2.4 Wave Energy Dissipation

The mechanical power extracted from the fluid per unit volume by the vegetation is calculated from the hydrodynamic force and the fluid velocity:

$$p_M = f_M^{vol} u \tag{2.8}$$

where:

- p_M is the instantaneous mechanical power extracted from the fluid per unit volume (W/m^3).
- f_M^{vol} is the force exerted by the fluid on the vegetation per unit volume (N/m^3).
- u is the horizontal fluid velocity (m/s).

The mean power extracted over one wave period is given by:

$$\overline{p_M}(x,z) = \frac{1}{T}\int_0^T f_M^{vol}(x,z,t)\, u(x,z,t)\, dt \tag{2.9}$$

The force f_M^{vol} is composed of two terms: drag and inertia. The mean contribution of the inertia term is zero for a periodic velocity field and a constant inertia coefficient, since:

$$\overline{u\frac{\partial u}{\partial t}} = \frac{1}{T}\int_0^T u\frac{\partial u}{\partial t}\, dt = \frac{1}{2T}\left[u^2\right]_0^T = 0 \tag{2.10}$$

Therefore, the mean power dissipated by the vegetation comes only from the drag contribution and is equal to:

$$\overline{p_D}(x,z) = \frac{1}{2}\rho_w C_D n d \overline{|u(x,z,t)|^3} \tag{2.11}$$

By integrating over the vegetation height, we obtain the vegetation-induced dissipation per unit horizontal area:

$$D_v(x) = \frac{1}{T}\int_{-h}^{-h+h_v}\int_0^T \frac{1}{2}\rho_w C_D n d |u(x,z,t)|^3 \, dt \, dz \tag{2.12}$$

where $D_v(x)$ is the mean wave energy dissipation rate per unit horizontal area ($\mathrm{W/m^2}$).

2.5 Wave Energy Attenuation

To compute wave attenuation, we consider the energy flux over a small horizontal element dx and account for the energy dissipated by the vegetation.

From linear wave theory, the mean wave energy per unit horizontal area is:

$$E(x) = \frac{1}{8}\rho_w g H(x)^2 \tag{2.13}$$

where $E(x)$ is expressed in $\mathrm{J/m^2}$.

The wave energy flux per unit width is defined as:

$$F(x) = E(x) c_g \tag{2.14}$$

where c_g is the group velocity, given by:

$$c_g = \frac{\omega}{2k}\left(1 + \frac{2kh}{\sinh(2kh)}\right) \tag{2.15}$$

The energy evolution along the horizontal direction is then obtained from the energy balance:

$$F(x) - F(x+dx) = D_v(x)\, dx \tag{2.16}$$

Taking the limit as dx approaches zero gives:

$$\frac{d(Ec_g)}{dx} = -D_v(x) \tag{2.17}$$

Since the water depth and wave period are constant, the group velocity c_g is also constant. Substituting Eq. (2.13) gives:

$$\frac{1}{8}\rho_w g c_g \frac{d(H^2)}{dx} = -D_v(x) \tag{2.18}$$

This finally gives the differential equation governing the evolution of the wave height:

$$\boxed{\frac{dH}{dx} = -\frac{4D_v(x,H)}{\rho_w g c_g H(x)}} \tag{2.19}$$

The equation is solved numerically using the `solve_ivp` function from `scipy.integrate`, with the initial condition:

$$H(0) = H_0 \tag{2.20}$$

At each integration step, the local wave height is used to update the orbital velocities and the corresponding energy dissipation. The numerical solution provides the evolution of wave height $H(x)$ along the vegetated area.

2.6 Results

The evolution of the wave height is shown in Fig. **??**.

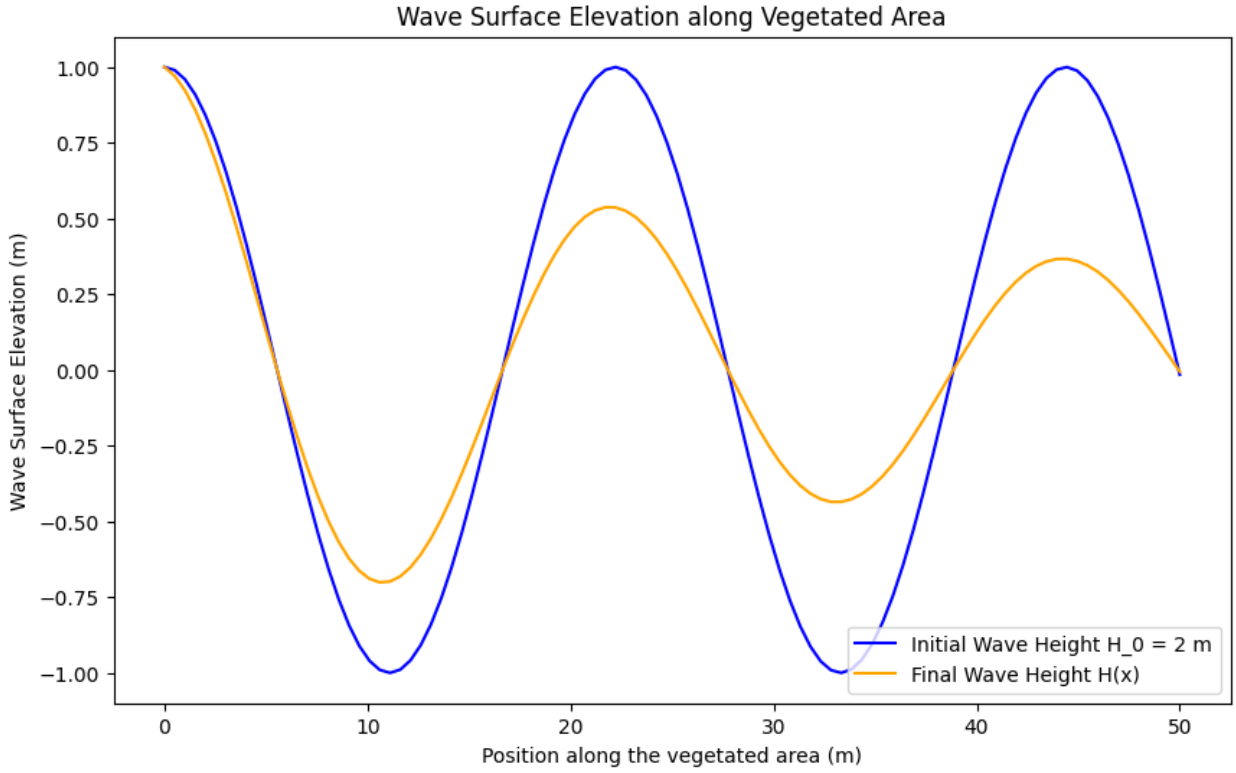


Figure 1: Wave Height evolution.

The wave height decreases progressively inside the vegetated region because the drag force continuously extracts energy from the wave motion. Once the wave leaves the vegetated region, the dissipation is set to zero and the wave height remains constant. This behaviour is consistent with the energy-balance formulation used in the model.

3 Second Model: Wave Attenuation through Rigid Cylinders under Irregular Waves

3.1 Main Assumptions

The regular wave model developed previously is extended to an irregular sea state. The vegetation condition remains the same: the stems are represented as rigid, fixed vertical cylinders.

The irregular surface elevation is represented in 2D by a superposition of linear wave components. Each sub-wave has its own frequency, wave number, phase and amplitude.

The following assumptions are made:

- Wave components propagate in the x direction
- Linear wave theory is applied for the fluid
- The sea state follows a JONSWAP spectrum
- The vegetation properties are identical to those used in the regular wave model.

3.2 JONSWAP spectrum

The incident irregular sea state is defined using a JONSWAP spectrum.

$$S(f) = \alpha \frac{g^2}{(2\pi)^4} f^{-5} \exp\left[-\frac{5}{4}\left(\frac{f_p}{f}\right)^4\right] \gamma^{\exp\left[-\frac{(f-f_p)^2}{2\sigma^2 f_p^2}\right]}, \tag{3.1}$$

where:

- $S(f)$ is the wave-energy spectral density (m^2/Hz);
- f is the wave frequency (Hz);
- $f_p = 1/T_p$ is the peak frequency (Hz);
- T_p is the spectral peak period (s);
- g is the gravitational acceleration (m/s^2);
- γ is the peak-enhancement factor;
- α is a scaling coefficient used to recover the prescribed significant wave height;
- $\sigma = 0.07$ for $f \leq f_p$ and $\sigma = 0.09$ for $f > f_p$.

The scaling coefficient α is determined numerically so that the spectrum satisfies

$$H_s = 4\sqrt{\int_0^\infty S(f)\,df}, \tag{3.2}$$

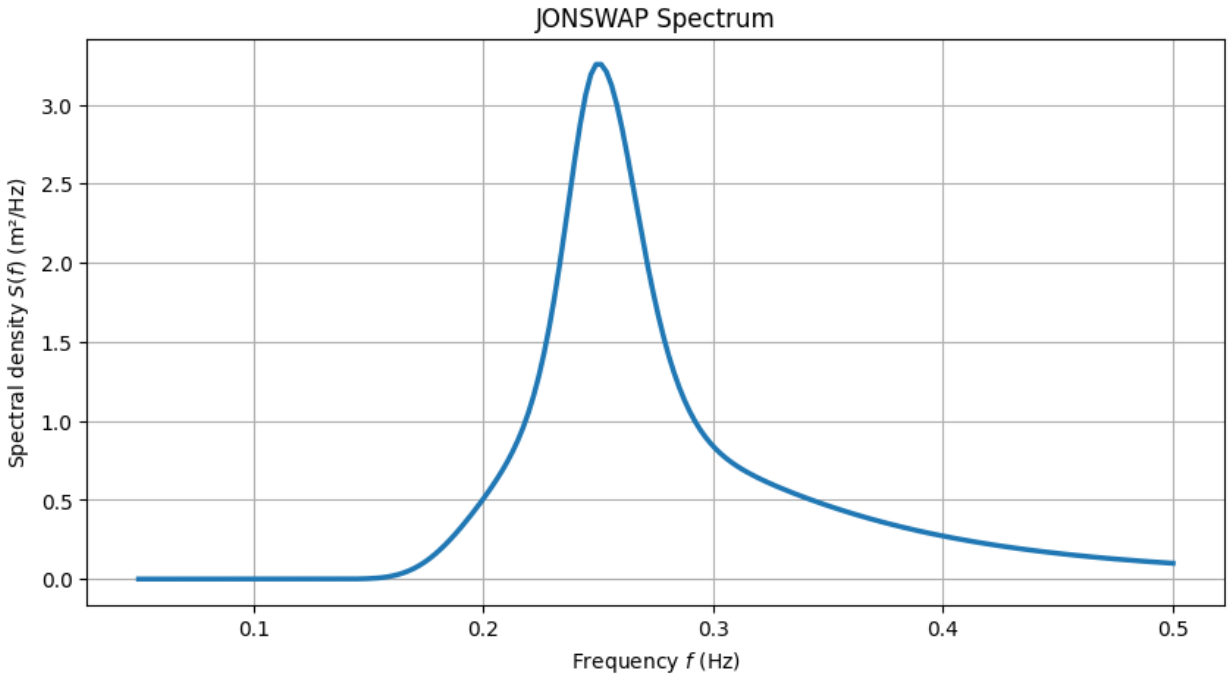


Figure 2: Jonswap Spectrum used in this study

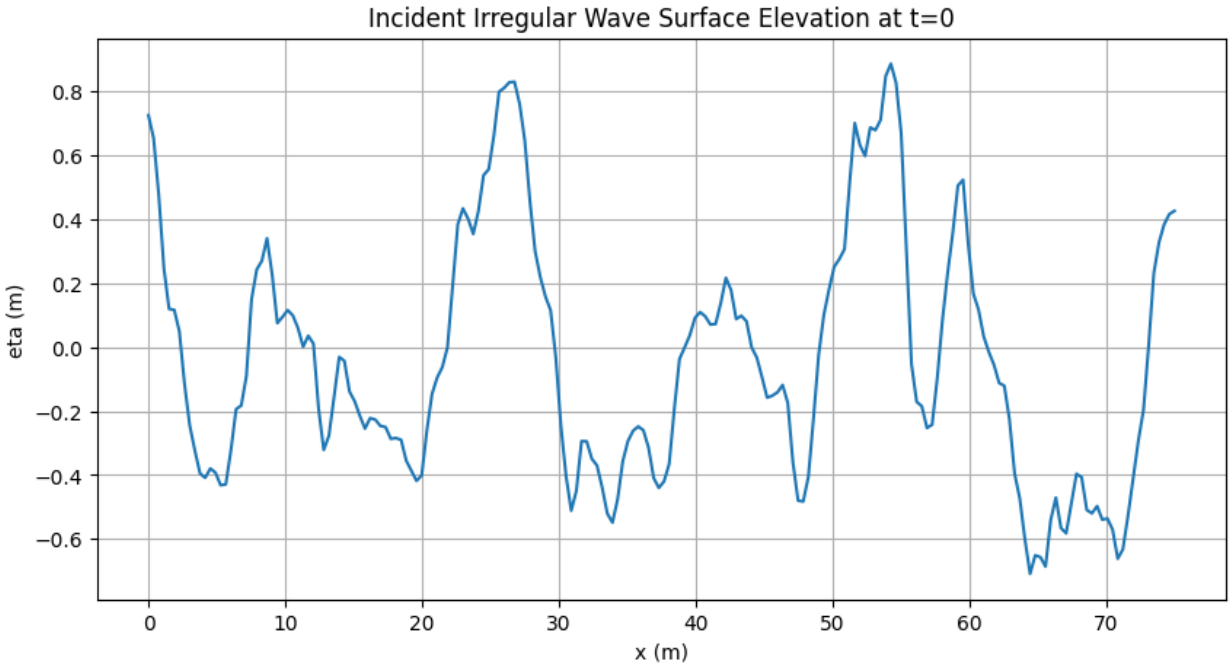


Figure 3: Free surface elevation at $t = 0.0$s

3.3 Spectral Discretisation

The frequency domain is divided into N discrete components separated by Δf. The amplitude of component j is:

$$a_j = \sqrt{2S(f_j)\Delta f}. \tag{3.3}$$

The angular frequency of component j is

$$\omega_j = 2\pi f_j. \tag{3.4}$$

For each component, the associated wave number k_j is obtained by solving the linear dispersion relation (eq 2.1). A random phase $\phi_j \in [0; 2\pi]$ uniformly distributed.

The irregular free surface elevation is then defined as follow:

$$\eta(x,t) = \sum_{j=1}^{N} a_j(x) \cos\left(k_j x - \omega_j t + \phi_j\right). \tag{3.5}$$

3.4 Velocity

The horizontal orbital velocity associated with frequency component j is

$$u_j(x,z,t) = a_j(x)\omega_j \frac{\cosh\left[k_j(z+h)\right]}{\sinh(k_j h)} \cos\left(k_j x - \omega_j t + \phi_j\right). \tag{3.6}$$

The total horizontal velocity is obtained by summing all frequency component:

$$u(x,z,t) = \sum_{j=1}^{N} u_j(x,z,t), \tag{3.7}$$

Because the drag force is nonlinear, it must be evaluated from the total reconstructed velocity at each x position rather than independently for each frequency component.

3.5 Distribution of the Drag Dissipation

The drag induced by the vegetation depends on the total horizontal velocity of the irregular wave. The calculation of the dissipation is made in two successive passes: we first pass evaluates the non linear drag force using the total velocity, the second pass uses the previous result to evaluate the dissipation for each component.

During the first pass, the total horizontal velocity defined by Eq. (3.7) is reconstructed by summing the contributions of all frequency components. The drag force is then calculated from this total velocity:

$$f_D^{vol}(x,z,t) = \frac{1}{2}\rho_w C_D n d |u(x,z,t)| u(x,z,t). \tag{3.8}$$

During the second pass, each frequency component is considered again. The total drag force calculated during the first pass is multiplied by the velocity u_j of component j. The resulting mean power contribution is integrated over the vegetation height and the duration of the time record.

$$D_j(x) = \frac{1}{T_r}\int_0^{T_r}\int_{-h}^{-h+h_v} f_D^{vol}(x,z,t) u_j(x,z,t)\, dz\, dt. \tag{3.9}$$

where T_r is the duration of the time record used to evaluate the mean dissipation.

The resulting value $D_j(x)$ is then used to update the energy flux and the amplitude of component j before advancing to the next horizontal slice. Repeating this procedure for every component allows the complete irregular wave spectrum to evolve progressively through the vegetated region.

3.6 Spatial Evolution of the Spectral Components

Now that the dissipation associated with each component has been evaluated, the energy flux of the component j at the position x_i is calculated as

$$F_j(x_i) = \frac{1}{2}\rho_w g a_j(x_i)^2 c_{g,j}, \tag{3.10}$$

where $a_j(x_i)$ is the local component amplitude and $c_{g,j}$ is its group velocity. The group velocity associated with component j is

$$c_{g,j} = \frac{\omega_j}{2k_j}\left(1 + \frac{2k_j h}{\sinh(2k_j h)}\right). \tag{3.11}$$

the energy dissipated while the component crosses the horizontal slice of length Δx is approximated by $D_j(x_i)\Delta x$. The energy flux at the position x_{i+1} is

$$F_j(x_{i+1}) = \max\left[F_j(x_i) - D_j(x_i)\Delta x, 0\right]. \tag{3.12}$$

The corresponding amplitude at the next position is obtained by inverting the energy flux relation

$$a_j(x_{i+1}) = \sqrt{\frac{2F_j(x_{i+1})}{\rho_w g c_{g,j}}}. \tag{3.13}$$

Now that we know the amplitude for each frequency f_j and position x_i, the free surface elevation is reconstructed as

$$\eta(x_i, t) = \sum_{j=1}^{N} a_j(x_i)\cos\left(k_j x_i - \omega_j t + \phi_j\right). \tag{3.14}$$

The local significant wave height is estimated from the standard deviation of the reconstructed surface elevation.

$$H_s(x_i) = 4\,\mathrm{std}\left[\eta(x_i, t)\right]. \tag{3.15}$$

3.7 Results

The parameters used in this model are:

Table 1: Numerical parameters used for the irregular-wave model.

Parameter	Symbol	Value
Significant wave height	H_s	2.0 m
Peak period	T_p	4.0 s
JONSWAP peak factor	γ	3.3
Number of components	N	100
Time-record duration	T_r	80 s
Vegetated length	L_v	50 m

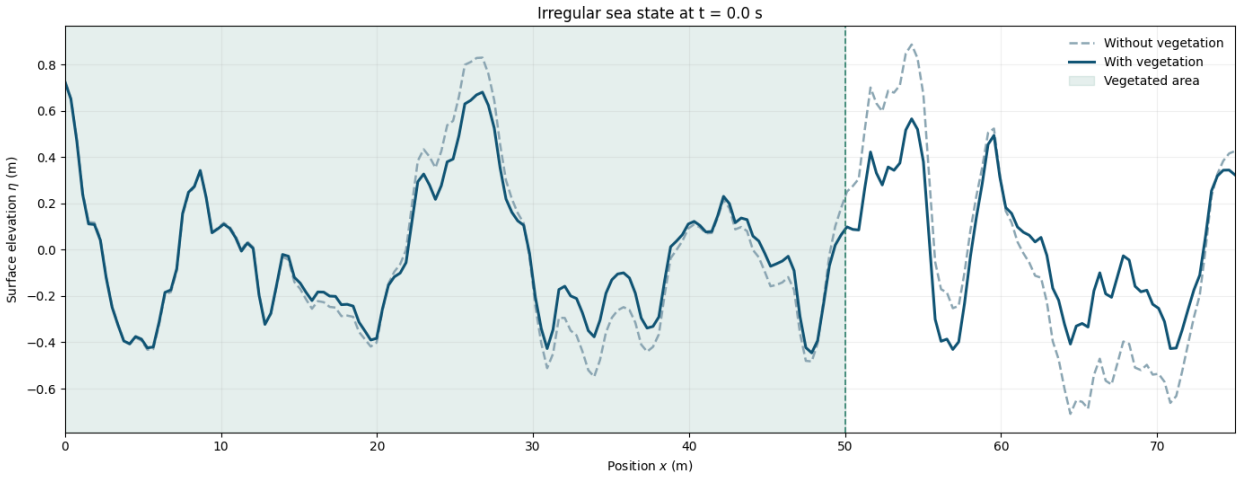


Figure 4: Evolution of the surface elevation

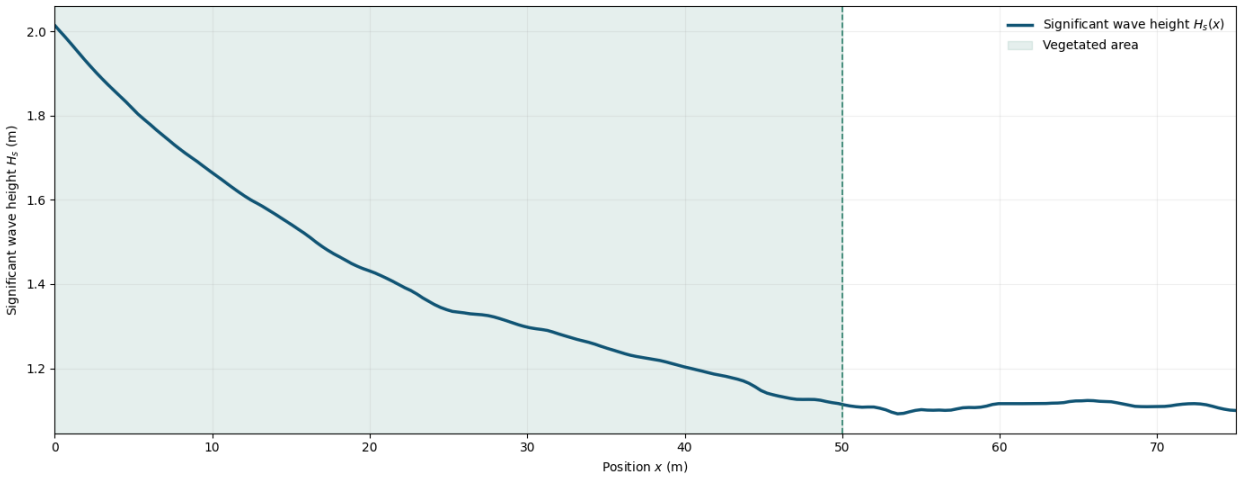


Figure 5: Evolution of the significant wave height

The comparison of the reconstructed free surface with and without vegetation shows that the same frequencies and phases are retained, while the amplitudes are progressively reduced inside the vegetated region. The corresponding significant wave height decreases along the vegetation and becomes constant downstream, where no additional dissipation is applied.